Cyber Threat Intelligence Report

Generated: 2026-10-10 14:44

Subject: Analyze the ransomware incident in the sample report.

Overall risk: High (8.5/10), likelihood Possible, P2 - High

1. Executive Summary

Threat type: Ransomware. Malware: Ransomware (generic). Attack summary: The attacker sent a phishing email with a malicious attachment to finance staff. Opening the attachment executed PowerShell, which downloaded payload files from 203.0.113.45 (evil-updates.example.com). The actor used stolen credentials, dumped LSASS memory, disabled Windows Defender, deleted shadow copies, and then encrypted files across the domain, halting billing for four days. MITRE techniques: T1566.001 Phishing: Spearphishing Attachment, T1059.001 PowerShell, T1078 Valid Accounts, T1562.001 Impair Defenses: Disable or Modify Tools, T1003.001 OS Credential Dumping: LSASS Memory, T1105 Ingress Tool Transfer, T1486 Data Encrypted for Impact, T1490 Inhibit System Recovery. Risk: High (8.5/10), likelihood Possible, priority P2 - High. Business impact: Operational disruption (billing halted for four days); Financial loss (average $3.7?M per ransomware incident). Response urgency: Act within 24 hours. Top immediate actions: Quarantine the compromised file servers and domain hosts from the network to stop further ransomware spread - Containment prevents encryption from reaching additional systems (T1486, T1490); Disable compromised user accounts and force password resets with MFA for all privileged accounts - Stops the attacker from using stolen credentials (Valid Accounts) (T1078); Terminate malicious PowerShell processes and delete the payload files downloaded from 203.0.113.45 (evil-updates.example.com) - Eradicates the execution vector and removes the tool transfer component (T1059.001, T1105).

2. Threat Overview

Threat type: Ransomware

Malware: Ransomware (generic)

Attack summary

The attacker sent a phishing email with a malicious attachment to finance staff. Opening the attachment executed PowerShell, which downloaded payload files from 203.0.113.45 (evil-updates.example.com). The actor used stolen credentials, dumped LSASS memory, disabled Windows Defender, deleted shadow copies, and then encrypted files across the domain, halting billing for four days.

Indicators of compromise

• 203.0.113.45

• evil-updates.example.com

Affected systems

• File servers

• Domain hosts

3. MITRE ATT&CK Mapping

Attack flow: Initial Access -> Execution -> Defense Evasion -> Credential Access -> Command and Control -> Impact

ID	Technique	Tactic	Confidence
T1566.001	Phishing: Spearphishing Attachment	Initial Access	0.95

ID	Technique	Tactic	Confidence
T1059.001	PowerShell	Execution	0.95
T1078	Valid Accounts	Defense Evasion	0.95
T1562.001	Impair Defenses: Disable or Modify Tools	Defense Evasion	0.95
T1003.001	OS Credential Dumping: LSASS Memory	Credential Access	0.95
T1105	Ingress Tool Transfer	Command and Control	0.95
T1486	Data Encrypted for Impact	Impact	0.95
T1490	Inhibit System Recovery	Impact	0.95

4. Risk Assessment

Severity: High (8.5/10)

Likelihood: Possible

Priority: P2 - High

Business impact

• Operational disruption (billing halted for four days)

• Financial loss (average $3.7?M per ransomware incident)

Affected assets

• File servers

• Domain hosts

Justification

The attack used multiple ATT&CK techniques to gain access, evade defenses, and encrypt data, stopping billing for four days and deleting recovery points, which shows high operational impact. Industry data cites an average $3.7?M financial loss per ransomware event, reinforcing a strong business impact. While the incident is severe, it does not affect the entire organization network, so the severity is set at 8.5. CISA reports a 10% annual chance of ransomware, placing the likelihood at Possible.

5. Mitigation Plan

Response urgency: Act within 24 hours.

Immediate actions

• Quarantine the compromised file servers and domain hosts from the network to stop further ransomware spread - Containment prevents encryption from reaching additional systems (T1486, T1490)

• Disable compromised user accounts and force password resets with MFA for all privileged accounts - Stops the attacker from using stolen credentials (Valid Accounts) (T1078)

• Terminate malicious PowerShell processes and delete the payload files downloaded from 203.0.113.45 (evil-updates.example.com) - Eradicates the execution vector and removes the tool transfer component (T1059.001, T1105)

• Re-enable Windows Defender, verify its configuration, and audit for tampering events - Restores endpoint protection that was disabled by the attacker (T1562.001)

• Detect and stop any LSASS memory dumping tools; ensure no further credential dumping occurs - Prevents additional credential theft (OS Credential Dumping) (T1003.001)

• Recreate system restore points after cleaning and, if possible, restore deleted shadow copies from offline backups - Mitigates the impact of Inhibit System Recovery technique (T1490)

Prevention

• Deploy multi-factor authentication for all user and privileged accounts - Reduces risk of credential reuse (Valid Accounts) (T1078)

• Enforce PowerShell constrained language mode and enable Script Block Logging (Event ID 4104) - Limits malicious PowerShell execution and provides forensic data (T1059.001)

• Implement email security gateway that scans attachments and conduct regular user training on phishing awareness - Blocks spearphishing attachment attempts (Phishing) (T1566.001)

• Configure network intrusion prevention to block outbound traffic to known malicious IPs such as 203.0.113.45 - Stops ingress tool transfer from the attacker’s C2 server (T1105)

• Restrict file and registry permissions to prevent disabling or modifying Windows Defender - Prevents Impair Defenses technique (T1562.001)

• Maintain immutable, offline backups and regularly test restore procedures - Ensures recovery capability despite shadow-copy deletion (Data Encrypted for Impact, Inhibit Recovery) (T1486, T1490)

Detection

• Enable PowerShell Script Block Logging and alert on encoded commands or download-cradle patterns - Detects malicious PowerShell activity (T1059.001) (T1059.001)

• Monitor for anomalous logins (geography, time, dormant accounts) and privileged account usage - Detects abuse of valid accounts (T1078) (T1078)

• Deploy Sysmon to generate alerts when non-system processes open lsass.exe or create MiniDump files - Detects LSASS memory dumping (T1003.001) (T1003.001)

• Alert on Windows Defender service stops or configuration changes via security-tool tamper events - Detects impairment of defenses (T1562.001) (T1562.001)

• Watch for mass file modifications, new encrypted extensions, and ransom note creation across file shares - Detects ransomware encryption activity (T1486) (T1486)

• Monitor execution of vssadmin, wmic, and bcdedit commands that delete shadow copies - Detects Inhibit System Recovery attempts (T1490) (T1490)

Recovery

• Restore encrypted data from verified, offline backups after confirming the environment is clean - Reinstates business operations without paying ransom

• Re-image compromised systems from known good baselines and apply latest patches - Ensures no residual malicious artifacts remain

• Conduct a thorough forensic investigation to identify any persistence mechanisms or lateral movement paths - Improves future detection and response

• Update incident response playbooks with lessons learned from this attack - Strengthens organizational readiness

• Notify affected stakeholders, regulatory bodies, and law enforcement as required - Meets compliance and facilitates coordinated response

ATT&CK mitigation reference

Technique	ATT&CK mitigations	Detection guidance
T1566.001 Phishing: Spearphishing Attachment	M1049 Antivirus/Antimalware, M1031 Network Intrusion Prevention, M1021 Restrict Web-Based Content, M1054 Software Configuration, M1017 User Training	Alert on emails carrying macro-enabled or executable attachments and on Office applications spawning script interpreters.

Technique	ATT&CK mitigations	Detection guidance
T1059.001 PowerShell	M1049 Antivirus/Antimalware, M1045 Code Signing, M1042 Disable or Remove Feature or Program, M1038 Execution Prevention, M1026 Privileged Account Management, M1021 Restrict Web-Based Content	Enable PowerShell Script Block Logging (Event ID 4104) and alert on encoded commands or download cradles.
T1078 Valid Accounts	M1032 Multi-factor Authentication, M1027 Password Policies, M1026 Privileged Account Management, M1018 User Account Management, M1017 User Training	Alert on logins from unusual locations or times and on use of dormant or privileged accounts.
T1562.001 Impair Defenses: Disable or Modify Tools	M1022 Restrict File and Directory Permissions, M1024 Restrict Registry Permissions, M1018 User Account Management, M1038 Execution Prevention, M1047 Audit	Alert on security-tool service stops, Defender configuration changes and tamper events.
T1003.001 OS Credential Dumping: LSASS Memory	M1040 Behavior Prevention on Endpoint, M1043 Credential Access Protection, M1025 Privileged Process Integrity, M1026 Privileged Account Management	Alert on non-system processes opening lsass.exe (Sysmon Event ID 10) and on comsvcs.dll MiniDump usage.
T1105 Ingress Tool Transfer	M1031 Network Intrusion Prevention	Monitor file downloads by certutil, bitsadmin, curl and PowerShell, and file creation from unusual external sources.
T1486 Data Encrypted for Impact	M1053 Data Backup, M1040 Behavior Prevention on Endpoint	Alert on mass file modification or renaming, new file extensions and ransom notes; deploy canary files.
T1490 Inhibit System Recovery	M1053 Data Backup, M1028 Operating System Configuration, M1038 Execution Prevention, M1018 User Account Management	Alert on vssadmin, wmic and bcdedit commands that delete shadow copies or disable recovery.

6. Evidence and Sources

Cited by the agents

• sample_threat_report.pdf, page 1 - billing halted for four days; file servers and domain encrypted; shadow copies deleted

• cisa-vulnerability-review-fy-2024-2025.pdf, page 3 - average financial loss $3.7M per ransomware incident; 10% chance of ransomware per year

• 260818.pdf, page 18 - Deletion of shadow copies and encryption of files with AES-256

7. Limitations and Assumptions

• The average $3.7?M loss applies to this organization despite lack of specific financial data

• Severity adjustment from baseline 10.0 to 8.5 reflects that impact, while severe, is limited to specific systems rather than the entire enterprise

• Network intrusion prevention can be configured to block the specific IP 203.0.113.45

• Immutable offline backups are available and have been tested

• Multi-factor authentication is not already enforced for all accounts

• PowerShell constrained language mode is supported in the environment

• Email security gateway with attachment scanning can be deployed promptly

• This report was generated automatically. Findings should be reviewed by a security analyst before action is taken.

Appendix A: Analysis Plan

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THREAT TYPE:
Ransomware

INFORMATION NEEDED:
- Detailed description of the incident (timeline, affected systems, ransom demand, and response actions)
- Malware characteristics (name, version, file hashes, execution method, persistence mechanisms)
- Indicators of compromise (IP addresses, domains, file names, registry keys, network traffic patterns)

AGENTS TO ACTIVATE:
1. Threat Analyzer — to dissect the incident, identify malware behavior, attack vectors, and relevant
indicators
2. MITRE Mapper — to map the identified attacker behavior to MITRE ATT&CK techniques, tactics, and
procedures

RETRIEVAL PLAN:

STEP 1:
Agent: Threat Analyzer
Retrieve: Full incident narrative, malware artifacts (hashes, file names, execution details), and any
documented indicators of compromise
Search query: “ransomware incident report details, malware hash, execution method, indicators of
compromise”
Purpose: Provide the foundational analysis of the ransomware event, including how it entered, spread, and
what it targeted

STEP 2:
Agent: MITRE Mapper
Retrieve: MITRE ATT&CK technique mapping for the behaviors identified by Threat Analyzer (e.g., initial
access, execution, persistence, exfiltration)
Search query: “ATT&CK mapping for ransomware behaviors, initial access, persistence, exfiltration
techniques”
Purpose: Translate the incident’s observed behaviors into standardized ATT&CK tactics and techniques for
contextual understanding

DEPENDENCIES:
- MITRE Mapper depends on the output of Threat Analyzer (the identified behaviors and indicators) to
accurately map to ATT&CK techniques.
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Appendix B: Agent Execution Status

Agent	Status
MITRE Mapper	success
Risk Assessment	success
Mitigation	success